

# Security Incident Report

Generated: 2026-08-09 18:03:24

Generated by AutoForensicAgent · Anomaly Detection & Digital Forensics Pipeline

UTC

Detected:

2026-08-09T18:03:04.606000

HIGH SEVERITY — Application Layer Protocol (T1071)

Anomaly Score: 0.628

## EXECUTIVE SUMMARY

Device **IOT-CAM-014** (IP Camera, Hikvision) at **192.168.1.47** was flagged for behavior consistent with **Application Layer Protocol** (MITRE ATT&CK T1071, tactic: Command and Control). The anomaly detection engine assigned a confidence score of **0.628**, and traffic evidence was automatically captured for forensic review under chain-of-custody ID **COC-ALERT-20260809-74A992**.

## DEVICE INFORMATION

|                       |                      |
|-----------------------|----------------------|
| DEVICE ID             | IOT-CAM-014          |
| DEVICE TYPE           | IP Camera            |
| VENDOR / FINGERPRINT  | Hikvision            |
| IP ADDRESS            | 192.168.1.47         |
| MAC ADDRESS           | 3C:EF:8C:11:9A:02    |
| FIRST SEEN ON NETWORK | 2026-05-02T08:12:00Z |

## DETECTION FEATURES

|                   |                    |
|-------------------|--------------------|
| PACKET RATE       | 1.5                |
| BYTE RATE         | 420.4              |
| UNIQUE DST IPS    | 1                  |
| UNIQUE DST PORTS  | 3                  |
| PORT ENTROPY      | 1.3382689280764646 |
| SYN RATIO         | 0.0                |
| UDP RATIO         | 0.0                |
| AVG PACKET SIZE   | 280.26666666666665 |
| FAILED AUTH COUNT | 0                  |
| NEW DST RATIO     | 1.0                |

## INCIDENT TIMELINE

- 2026-08-09T18:06:34.357666+00:00  
First packet observed from IOT-CAM-014
- 2026-08-09T18:06:43.690999+00:00  
Traffic window closed (15 packets analyzed)
- 2026-08-09T18:03:04.606963+00:00  
Anomaly detector flagged window as 'c2\_communication'

## MITRE ATT&CK MAPPING

T1071 – Application Layer Protocol

Command and Control

TACTIC: Command and Control

Adversaries may communicate using application layer protocols to avoid detection by blending in with existing traffic, often indicating the device has already been compromised and is receiving instructions from a remote server.

### Recommended Actions

- Immediately isolate the affected device from the network
- Block identified C2 IP/domain at the firewall
- Perform full firmware reset and forensic imaging before restoring device

## EVIDENCE & CHAIN OF CUSTODY

PCAP FILE: evidence/ALERT-20260809-74A992\_capture.pcap  
SHA-256: adae819c4bc9046621fc46a7aea97ca687e723c82d7fba22e7c2ff25e68d7d04  
CHAIN OF CUSTODY ID: COC-ALERT-20260809-74A992  
CAPTURED BY: NetGuard AutoForensicAgent v1.0

This report was generated automatically by the IoT Anomaly Detection & Digital Forensics module. It is intended to support, not replace, human security analyst review. Evidence integrity should be verified by re-computing the SHA-256 hash of the referenced pcap file before use in any formal investigation.

REPORT INTEGRITY HASH (SHA-256): 674f37b50e562e8ace9448199206713cd0844a35aa608bde2eec5e9a6a70bdbb